

WRANGELL, AK, USA

WMO: 703870

Lat: 56.483N

Lon: 132.367W

Elev: 43

StdP: 14.67

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25338

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	16.3	-2.3	4.9	18.4	0.9	5.8	19.4	25.7	38.1	24.3	35.7	8.9	80	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	71.5	60.0	67.6	58.0	64.2	56.1	60.7	68.9	59.0	65.3	57.6	63.1	6.6	310

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.8	69.0	62.8	55.3	65.3	61.0	54.6	63.5	60.2	26.8	69.4	25.6	65.3	24.8	63.0	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.6	19.5	17.2	9.6	77.6	4.0	3.2	6.7	79.9	4.4	81.8	2.2	83.6	-0.7	86.0		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			9.6	77.6	4.0	3.2	6.7	79.9	4.4	81.8	2.2	83.6	-0.7	86.0		
			7.4	63.7	3.5	1.5	4.9	64.8	2.9	65.6	0.9	66.5	-1.6	67.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	33.1	34.3	37.4	43.1	49.9	54.8	57.2	57.0	52.3	45.2	38.3	34.0	
	DBStd	10.39	8.68	6.74	5.42	3.97	4.64	4.20	3.24	3.37	3.26	4.15	5.72	7.42	
	HDD50	2647	525	440	392	212	59	3	0	0	13	155	351	497	
	HDD65	7388	990	860	857	658	468	307	243	248	381	613	801	962	
	CDD50	738	0	0	0	4	56	148	223	218	83	6	0	0	
	CDD65	4	0	0	0	0	0	2	1	1	0	0	0	0	
	CDH74	28	0	0	0	0	0	15	6	7	0	0	0	0	
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0	
Wind	WSAvg	6.5	8.0	7.5	7.8	6.9	5.8	5.0	4.0	4.2	5.4	7.2	8.1	8.6	
Precipitation	PrecAvg	79.1	7.3	5.4	5.1	4.7	4.4	4.2	4.6	5.6	8.9	13.1	8.4	7.5	
	PrecMax	95.7	14.6	9.8	9.7	8.3	7.4	6.7	8.9	12.9	17.0	23.4	17.5	12.8	
	PrecMin	64.3	1.6	0.0	1.5	0.8	1.7	1.5	1.4	1.4	1.7	7.7	1.7	1.9	
	PrecStd	8.6	3.2	2.7	2.0	1.9	1.6	1.3	1.9	2.7	3.5	4.0	4.1	2.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	48.2	53.7	63.1	70.4	75.3	74.7	73.3	65.8	57.3	52.1	49.4	
		MCWB	43.8	42.1	44.7	49.7	55.1	61.1	62.3	61.6	56.8	50.7	46.7	44.1	
	2%	DB	45.9	45.9	48.2	56.3	65.5	70.1	70.4	69.9	61.9	54.5	48.5	46.3	
		MCWB	42.2	41.1	42.3	46.1	53.3	58.4	60.5	59.7	54.5	50.2	45.5	42.9	
	5%	DB	44.9	44.8	45.8	52.6	62.2	65.7	66.0	66.0	59.2	52.4	46.4	45.1	
		MCWB	41.2	40.4	40.8	44.1	51.4	56.4	58.3	58.4	53.7	48.7	43.4	41.8	
	10%	DB	43.0	42.9	44.7	49.6	57.4	62.8	63.3	63.2	57.0	50.7	45.3	43.1	
		MCWB	39.6	39.1	39.9	43.2	49.2	54.7	56.7	56.8	52.6	47.3	42.0	39.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.1	43.4	45.7	50.4	57.0	62.0	63.0	62.5	58.8	53.3	48.2	46.0
			MCDB	47.1	46.6	50.6	60.7	67.6	74.2	72.9	72.2	63.3	55.5	50.4	47.8
2%		WB	43.0	41.6	43.4	47.0	54.3	59.0	60.8	60.4	56.4	50.9	46.1	43.8	
		MCDB	45.7	45.2	47.5	54.5	63.4	67.8	69.3	67.6	60.1	53.7	48.2	46.6	
5%		WB	41.3	40.7	41.6	45.4	52.3	57.0	58.9	59.0	54.9	49.2	44.3	41.6	
		MCDB	44.6	44.2	45.4	50.6	59.5	64.3	64.8	64.5	58.1	51.8	46.5	44.8	
10%		WB	40.0	39.6	40.4	43.8	50.7	55.2	57.4	57.6	53.4	47.5	42.3	40.1	
		MCDB	43.2	43.0	44.1	48.7	56.7	61.0	62.4	62.4	55.9	49.8	45.2	43.2	
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.0	8.4	10.9	12.1	10.9	9.7	10.1	8.6	6.8	6.1	6.6	
		MCDBR	7.5	8.3	11.1	17.4	19.7	19.1	17.5	17.4	12.9	9.1	7.5	7.6	
		MCWBR	5.6	5.6	7.2	9.6	10.3	10.2	9.5	9.8	7.6	6.0	5.5	5.5	
	5% WB	MCDBR	7.3	7.5	9.9	14.3	16.6	16.1	14.7	14.8	10.6	7.7	7.4	7.1	
		MCWBR	5.6	5.4	6.8	8.4	9.2	9.0	8.4	8.7	7.0	5.8	5.8	5.5	
Clear-Sky Solar Irradiance	taub	0.286	0.302	0.331	0.376	0.396	0.385	0.402	0.369	0.356	0.328	0.291	0.276		
	taud	2.211	2.298	2.240	2.143	2.169	2.322	2.326	2.451	2.455	2.413	2.229	2.137		
	Ebn at Noon	190	241	261	265	266	272	264	267	253	229	184	156		
	Edh at Noon	17	23	32	41	43	37	36	30	26	21	16	14		
All-Sky Solar Radiation	RadAvg	144	386	714	1172	1497	1506	1311	1123	728	395	177	107		
	RadStd	19	75	107	115	171	134	173	129	117	62	28	18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.77	+1.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	N/A

Nomenclature: See separate page